

EXP1-3 Bridge Design P13

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Section1-Introduction to bridge design:

Bridge is a kind of overhead building with carrying capacity, which carries the heavy responsibility of promoting political, economic and cultural exchanges.

In terms of transportation, bridges serve as access, extension and network functions. They cross the natural barrier and form an integrated transportation system, greatly reducing the commuting time, which reflects improvements of people's livelihood in a country. For instance, as the first bridge on Yangtze River, *Wuhan Yangtze River Bridge* connected the arteries between the North and the South in China, becoming a great achievement of new China's engineering construction [1].

In terms of economy, it is easy to form markets at both ends of bridges. Bridge can bring huge flows of traffic and people across rivers. With the movement, markets are created in near area. *Moreover, bridge with a long history can also be used as tourism resources to generate revenue for cities.* Landscape function cannot be ignored, its historical, cultural value and beautiful sights can be used as tourist attractions, and create tourism value.

In the aspect of and culture, bridge is the carrier of bridge culture, environmental aesthetics, bridge technology and the important object of literary and artistic creation. Embodying the cultural connotation of bridge is also one of its basic functions.

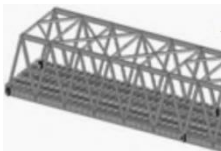
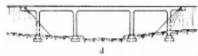
Types	Brief picture	Mechanical characteristics	Advantages	Disadvantages
Beam bridge		Mainly by bending, bearing no thrust.	Can be used in poor geological conditions.	There need to be piers under the deck, so it is hard to be used in our experiment.
Arch bridge		Under vertical load, bridge bearing horizontal thrust, this horizontal thrust significantly offset the load in the bearing structure arch ring or arch ribs caused by the bending moment.	The bridge span capacity is strong, which can bear vertical force	The bridge pier in this experiment is too small to place the arch bridge
Truss bridge		Under the vertical load, the beam and column have bending moment and axial force, and the column foot also produces horizontal thrust.	Suitable for three-dimensional cross line.	The amount of consumables is larger than the other five structures
Suspension bridge		Under firm load, the cable is subjected to great tension by derrick.	Cost less than other structure.	There need to be piers under the deck, so it is hard to be used in our experiment.
Cable-stayed bridge		The stay cables provide vertical support for the main beam, which has bending moment and axial force.	Cross the ability is very strong	There need to be piers under the deck, so it is hard to be used in our experiment.
Composite system bridge		A bridge consisting of several structures. The essence is to use different combinations of beam and arch to make the stress of the bridge more reasonable.	The structure can be adjusted according to the situation	The process is too complicated and difficult, it's hard to bring into our experiment.

Fig.1 The six most common bridge structures [2]

Section2-Our bridge design:

In this experiment, we are going to use paper straws and tapes to build a bridge, which can span a distance of 920 millimeters and can carry at least a kilogram. We plan to build a bridge by the structure of truss. Due

to the properties of paper straw and tape, it's hard to cross the gap and carry a weight of one kilogram only by simple splicing. However, the truss structure can make up for these defects on the whole, meeting the demand for safety and load bearing in principle. So, for the main part of the bridge, we use a double-clamped triangle model, which is composed of two lines of paper straws connected above and below, and a row of repeated triangles composed of three straws is sandwiched between the two lines of straws. This truss structure can ensure the stability and bearing capacity of the bridge body to the greatest extent, help the bridge to cover a distance of 920mm without piers.

Inspired by *Wuhan Yangtze River Bridge*, our bridge design principles are divided into three points. First is the stability, *The most stable principle of triangle* is used all over the bridge. By using positive triangle, inverted triangle, our structure makes up for the fragile characteristics of the material to the greatest extent. The second point is the economy, it's important to distribute the number and length of straws on each component, and the key structure need to be strengthened as possible. The last point is innovation, considering the particularity of this experiment, we need to make innovations in the design to better adapt to the current environment.

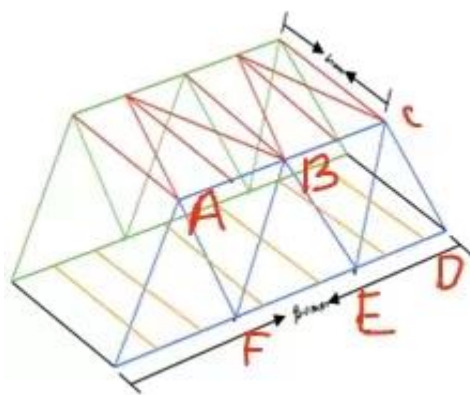


Fig.2 Brief picture of our bridge

In order to make the structure stiff enough, many forms are used to reinforce and connect bridge. Not only we bind paper straws with tape, but also bending and folding straws, insert straws head-to-head to connect are used to reinforce our bridge. Moreover, to adapt the special situation of this experiment, we create a new type of triangular stitching pieces, which can evenly distributed load bearing required force.

Diagram and special features of our bridge:

Part	Picture	Force analysis
A		Finally considering the A point, AB bar connected to A point stretched to the right, while the left side provide the equal force opposite to the previous force.
B		Point B is subjected to a downward pressure and a pull to the right. BF is required to provide an oblique force to the right and the AB rod is required to provide a horizontal pull to the left.
C		The BC bar connected to point C provides downward pressure, the oblique bar CE connected to point C compresses to provide oblique upward force, and the BC bar stretches to provide horizontal force to the right.
E		the EF rod needs to provide the force to the right because of the compression for the balance and the EB rod stretched to provide the vertical upward force.
F		In order to balance, the base at F point needs to provide upward force to resist the force down to the left provided by BF rod.

Part of bridge	Brief picture	Special features
Triangular component		Through the continuous arrangement of multiple triangular structures to obtain the overall strength of the bridge.
Triangular stitching pieces		Maximum splicing strength is obtained by habitually folding and binding tape at the joint of the straw.

Reference

1. Wuhan Yangtze River Bridge - Wikipedia [Internet]. En.wikipedia.org. 2022 [cited 19 October 2022].
2. Zhang Xinchun, Lu Zelei, Zhou Li, et al. Study on Stability of Steel Truss Structure during Flat Pushing Process [J]. Journal of Xi'an University of Science and Technology, 2022, 42(4) : 716-723.